| SUSPENSION—VEHICLE LEAN CORRECTION PROCEDURE | Article No. 02-18-1 |
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**FORD:** 2000-2003 F-650, F-750

## ISSUE

Some 2000-2003 F650/750 up to and including 30K GVW may exhibit a vehicle lean condition. This may be caused by unequal vehicle weight distribution, spring variation, frame assembly (twist, bow, or matchboxing), or unequal cab mounts/mounting. This correction procedure has been developed for diagnosing and correcting vehicle lean on these vehicles.

## ACTION

Many times the contributors to vehicle lean will settle out during initial driving in a loaded condition. Except for corrections made as a result of visual checks for bumper damage, frame damage and correct springs, spacers, tires, tire pressure and body mounts, make no repairs until vehicle has had a secondary body installed and has been driven in a loaded condition for 1,000 miles (1,600km).

Prior to any measurements taken, ensure that all tires are correctly inflated. The maximum allowable differential between left and right side of the vehicle is 1 inch (25.4mm) measured from outer edge of bumper to a level ground (Figure 1), and/or rear of cab at lower rocker panel (Figure 2). Correction should not be attempted on vehicles with 1 inch (25.4mm) or less difference between the extreme right and left sides.

**SERVICE PROCEDURE**

To determine the cause of the lean it will be necessary to place the vehicle on a flat level surface, such as an alignment bay. Then measure and record the distance between the top of the front axle to the frame on both sides (Figure 3).

**Low Profile Vehicles (19.5″ Tires):**

If the difference is between 1/2″ and 3/4″, it will be necessary to install a 1/2″ suspension spacer 3C4Z-5A300-AA on the low side of the vehicle. If the difference is greater than 3/4 ″, it will be necessary to install a 1″ suspension spacer F81Z-5A300-BA on the low side of the vehicle.

**NOTE**

**IF A 1″ AXLE SPACER IS NEEDED TO CORRECT THE LEAN ON A LOW-PROFILE VEHICLE WITH A FRONT STABILIZER BAR, A NEW, LONGER FRONT U-BOLT F81Z-5455-MB IS REQUIRED .**

**Dock Height Vehicles (22.5″ Tires):**

If the difference is greater than 1/2″, it will be necessary to remove the existing 1″ suspension spacer on the high side (note different than low profile vehicles) and install a 1/2 ″ suspension spacer 3C4Z-5A300-AA.

**NOTE**

**UNDER NO CIRCUMSTANCE SHOULD A DOCK HEIGHT VEHICLE BE WITHOUT AT LEAST A 1/2″ SUSPENSION SPACER UNDER EACH SIDE FOR CLEARANCE REQUIREMENTS.**

To remove and add suspension spacer, reference procedure in Section 204 of Workshop Manual. Note installation of the suspension spacer is to be under the shock mounting bracket and above the axle.

**CAUTION**

**PROFESSIONAL FRAME STRAIGHTENING SHOP PROCEDURES ARE NOT RECOMMENDED BECAUSE THEY MAY CAUSE FRAME RAIL OR CROSSMEMBER DAMAGE.**

Re-Cambering (or arching) of springs is also not recommended due to the reduced fatigue life of the spring.

Do not install more than one (1) suspension spacer per side to maintain steering performance criteria.

Only after insuring that the suspension is level, it may be necessary to adjust the cab. To determine if cab adjustment is required, measure top of frame rail vertically to lower flange at back of cab (Figure 4). If measurement difference from left to right is greater than 1/4″, a cab spacer 3C4Z-5D018-AA will be required on the low side front body mount.

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In order to install the new body insulator spacer, follow Workshop Manual Section 502-02 for removing the body mount retaining bolt. Install only one (1) spacer in between the body of the vehicle and the insulator. Follow Workshop Manual procedure to install the new body mount retaining bolt W705264-S301.

**CAUTION**
**TO MAINTAIN SUFFICIENT BOLT THREAD ENGAGEMENT, DO NOT INSTALL MORE THAN ONE (1) CAB SPACER.**

| PART NUMBER | PART NAME |
|---|---|
| 3C4Z-5A300-AA | 1/2″ Suspension Spacer |
| F81Z-5A300-BA | 1″ Suspension Spacer |
| F81Z-5455-MB | U-Bolt |
| 3C4Z-5D018-AA | Cab Mount Spacer |
| W705264-S301 | Body Mount Retaining Bolt |

**OTHER APPLICABLE ARTICLES:** NONE
**WARRANTY STATUS:** INFORMATION ONLY
**OASIS CODES:** 304000, 305000

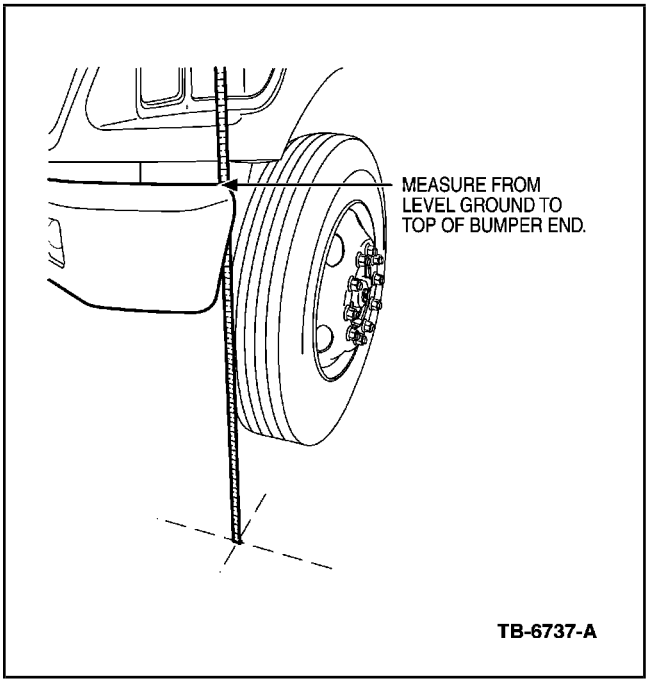



Figure 1 - Article 02-18-1

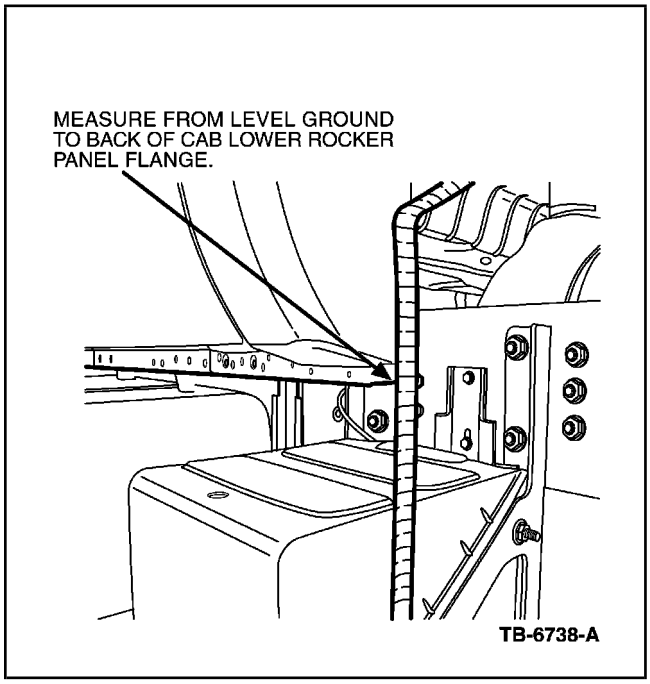



Figure 2 - Article 02-18-1

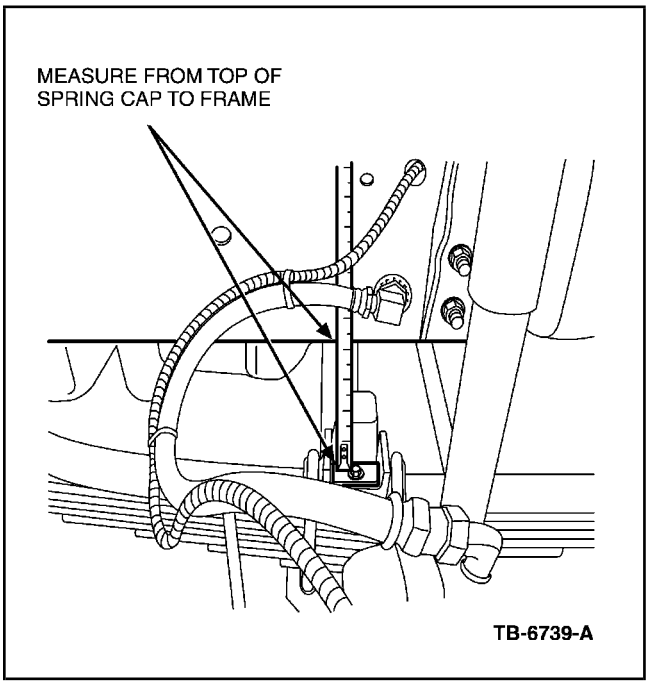



Figure 3 - Article 02-18-1

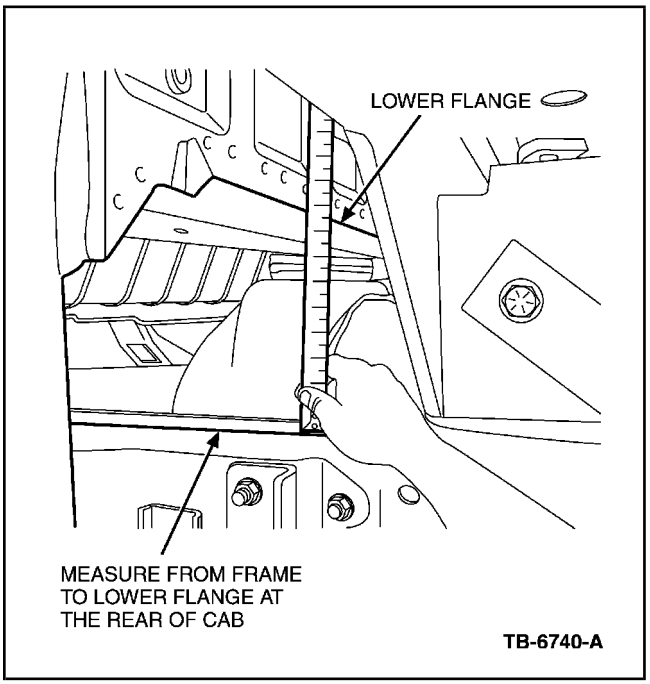



Figure 4 - Article 02-18-1

**NOTE:**The information in Technical Service Bulletins is intended for use by trained, professional technicians with the knowledge, tools, and equipment to do the job properly and safely. It informs these technicians of conditions that may occur on some vehicles, or provides information that could assist in proper vehicle service. The procedures should not be performed by "do-it-yourselfers". Do not assume that a condition described affects your car or truck. Contact a Ford, Lincoln, or Mercury dealership to determine whether the Bulletin applies to your vehicle.